

Instructions



Hello! How is it going?



amazing
hi lol great
good
sadlyy
awesome ok
hungry no much

Agenda

- Recall last session
- Discuss current topic
- Practice
- Quiz

Total Recall



What do you remember most from few previous workshops? (short answer)

unit testing

tdd, unittest

tdd

mokito

spy

junit, mockito

uni testing, mikito

What happens when an assumption fails?

7 ✓

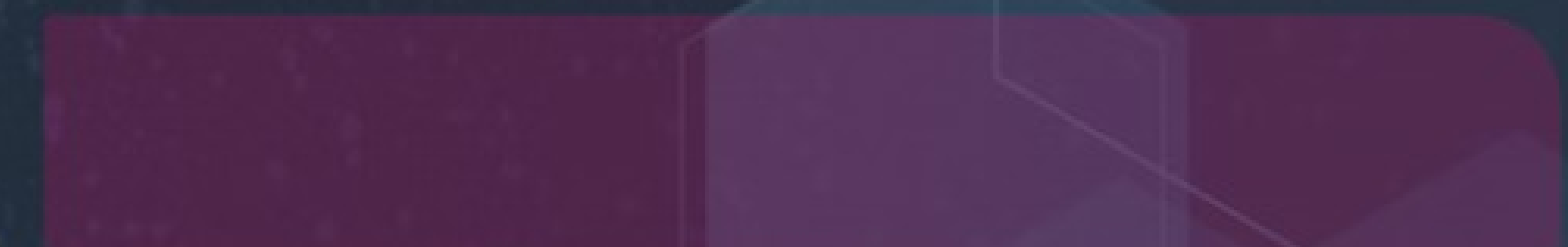


The test will be aborted.

0 ✗

The test will be ignored.

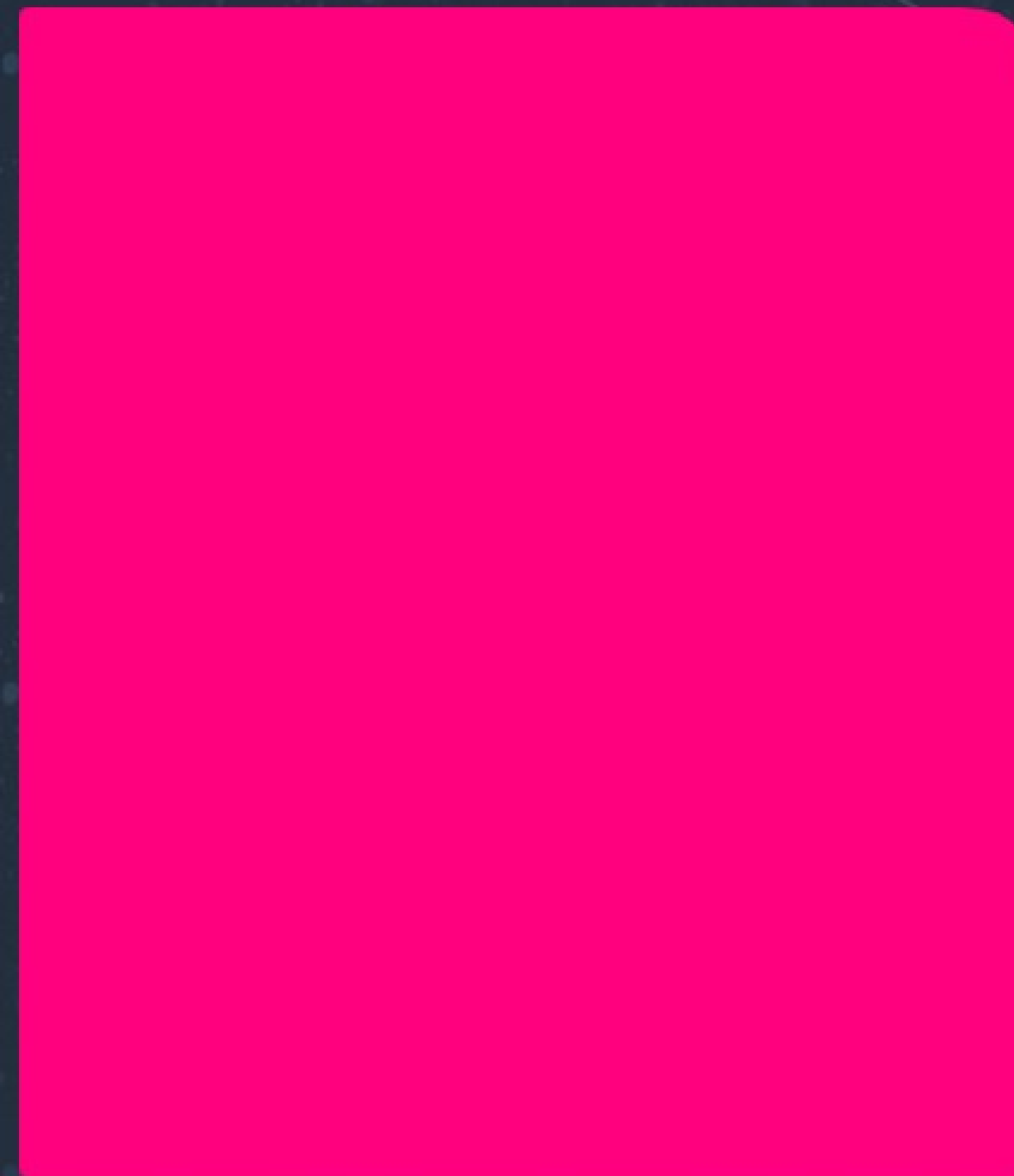
1 ✗



The test will fail.

What does @BeforeEach do in JUnit5?

12 ✓



0 ✗

Runs after all tests

0 ✗

Skips failing tests

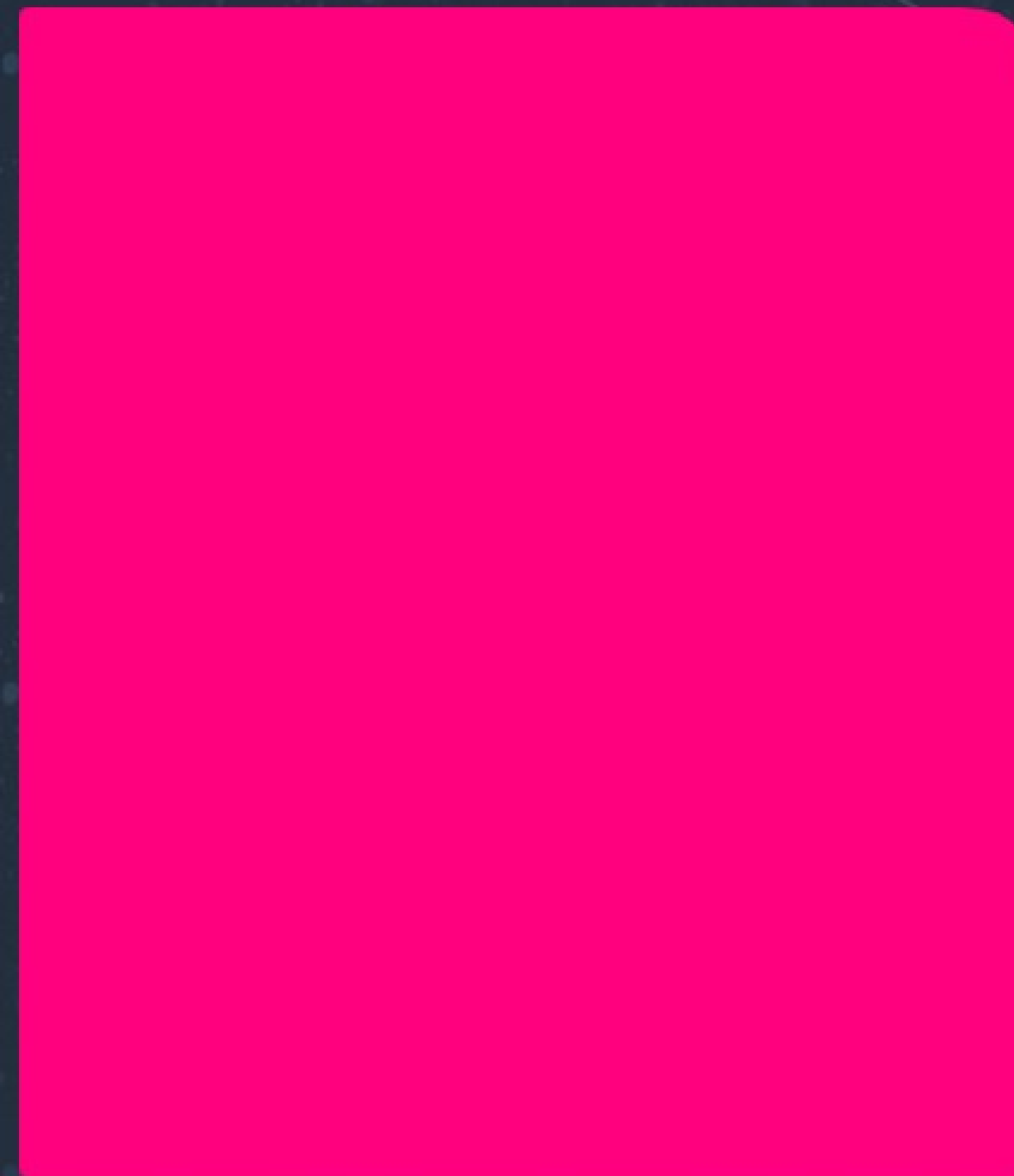
0 ✗

Runs before each test method

Marks a test as optional

Which JUnit method verifies an expected exception?

14 ✓



0 ✗

assertNotNull()

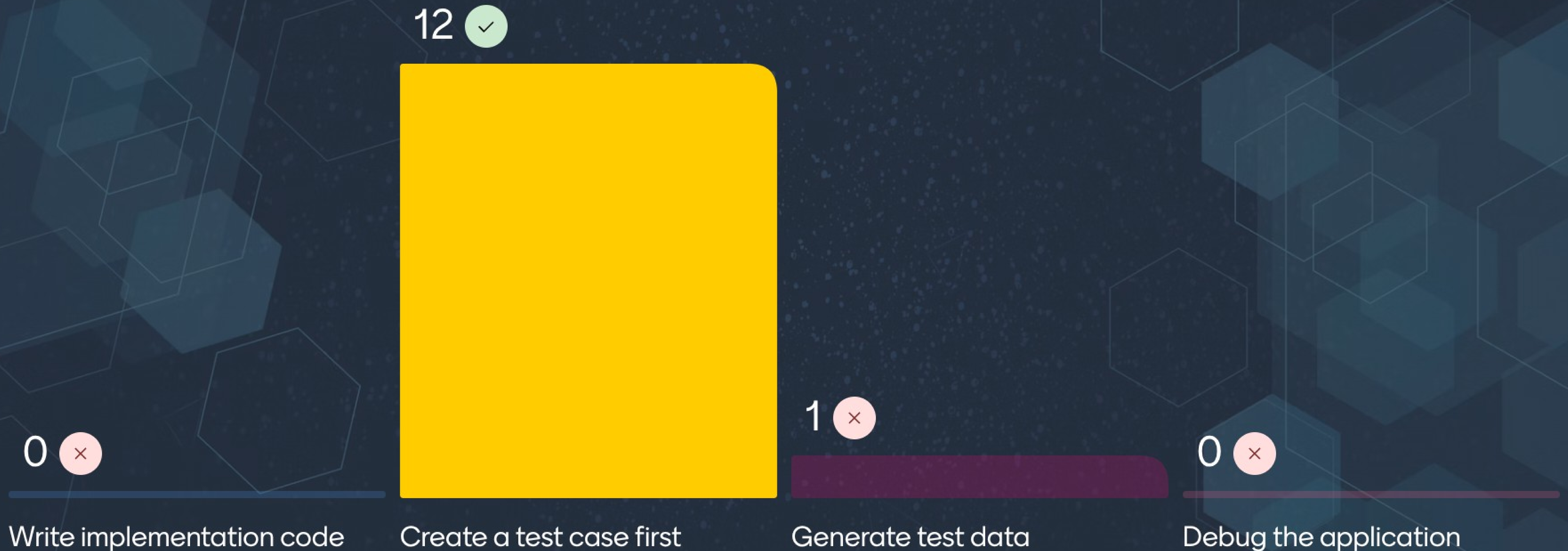
0 ✗

afterEach()

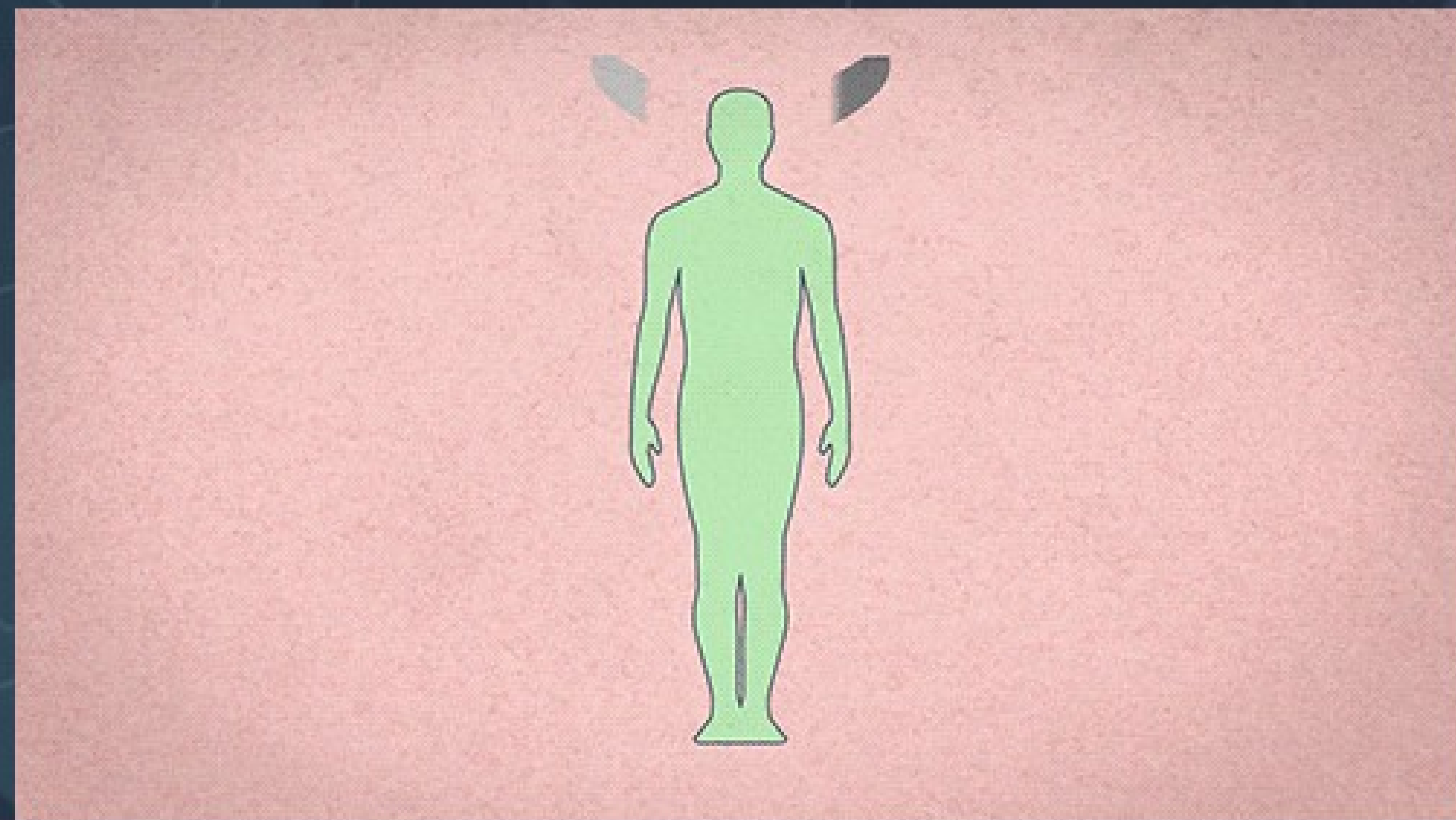
0 ✗

assertContains()

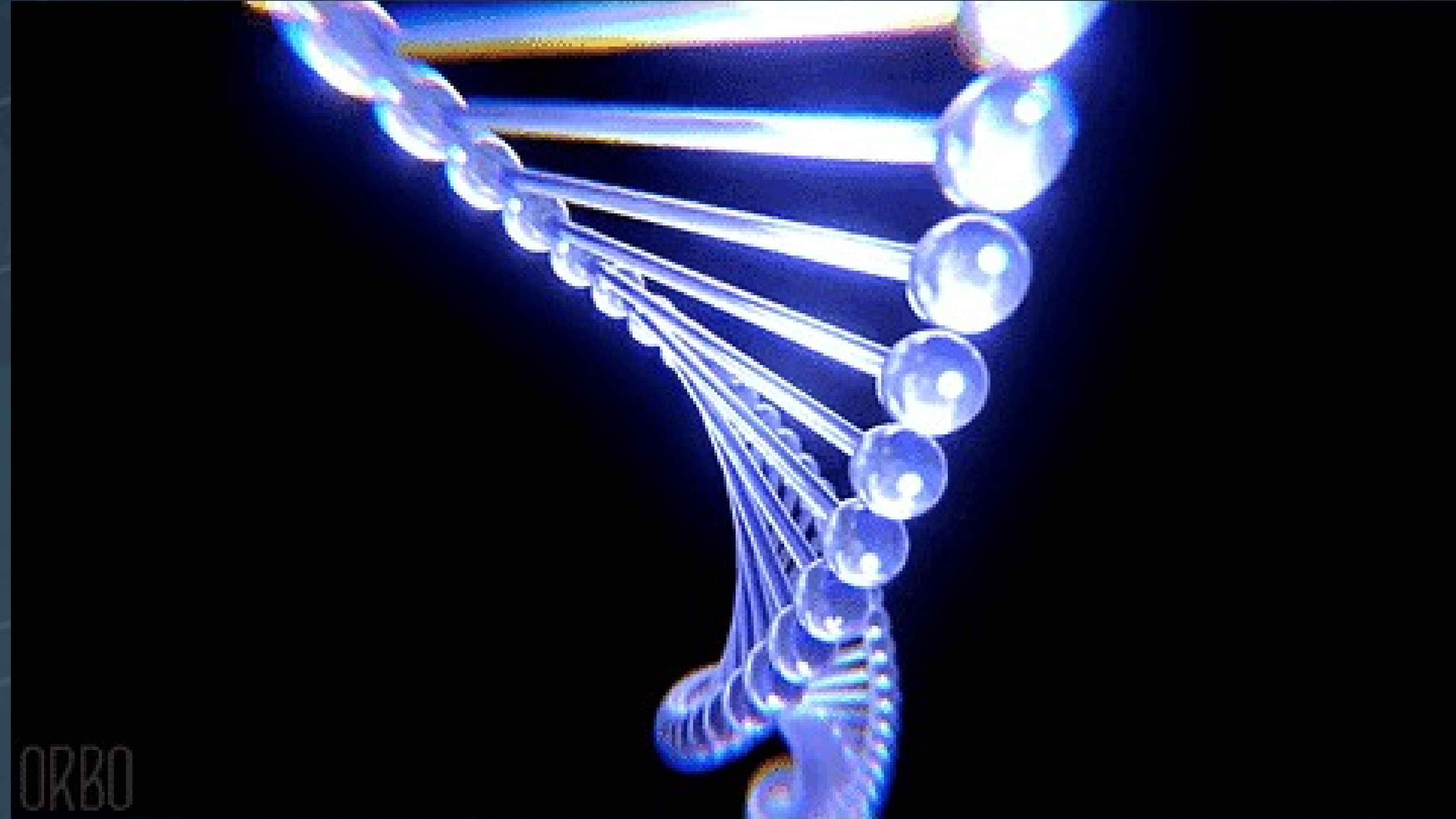
Example: What does TDD workflow start with?



ARRANGE -> ACT -> ASSERT



Next topic



Random fact

If you take all the DNA in your cells and stretch them out it can reach from the 🌍 to the 🌙 over 100,000 times.

Date Time

Timeline, Local, and Zone Time

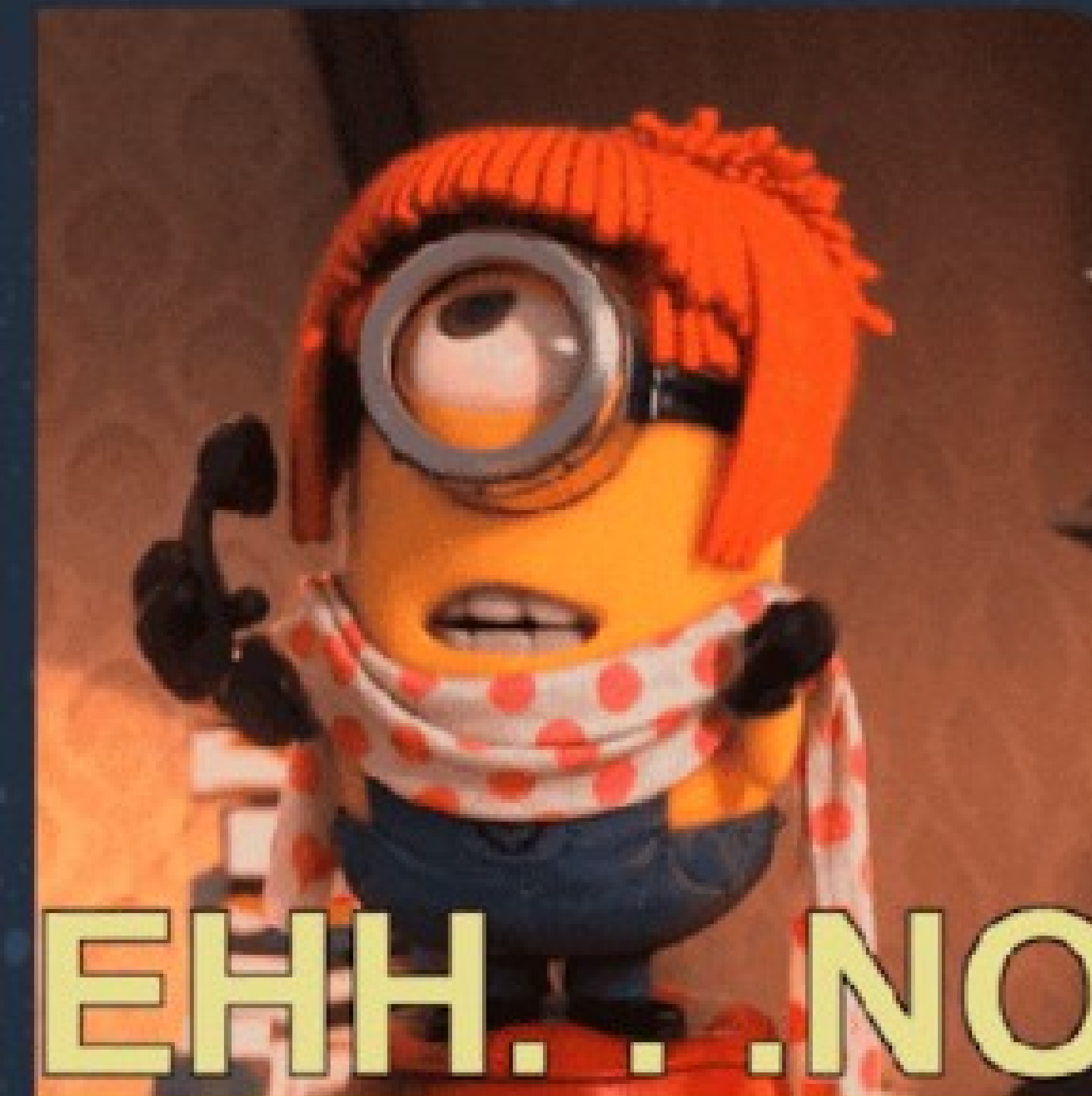
Have you looked through the materials?

5



yes

4



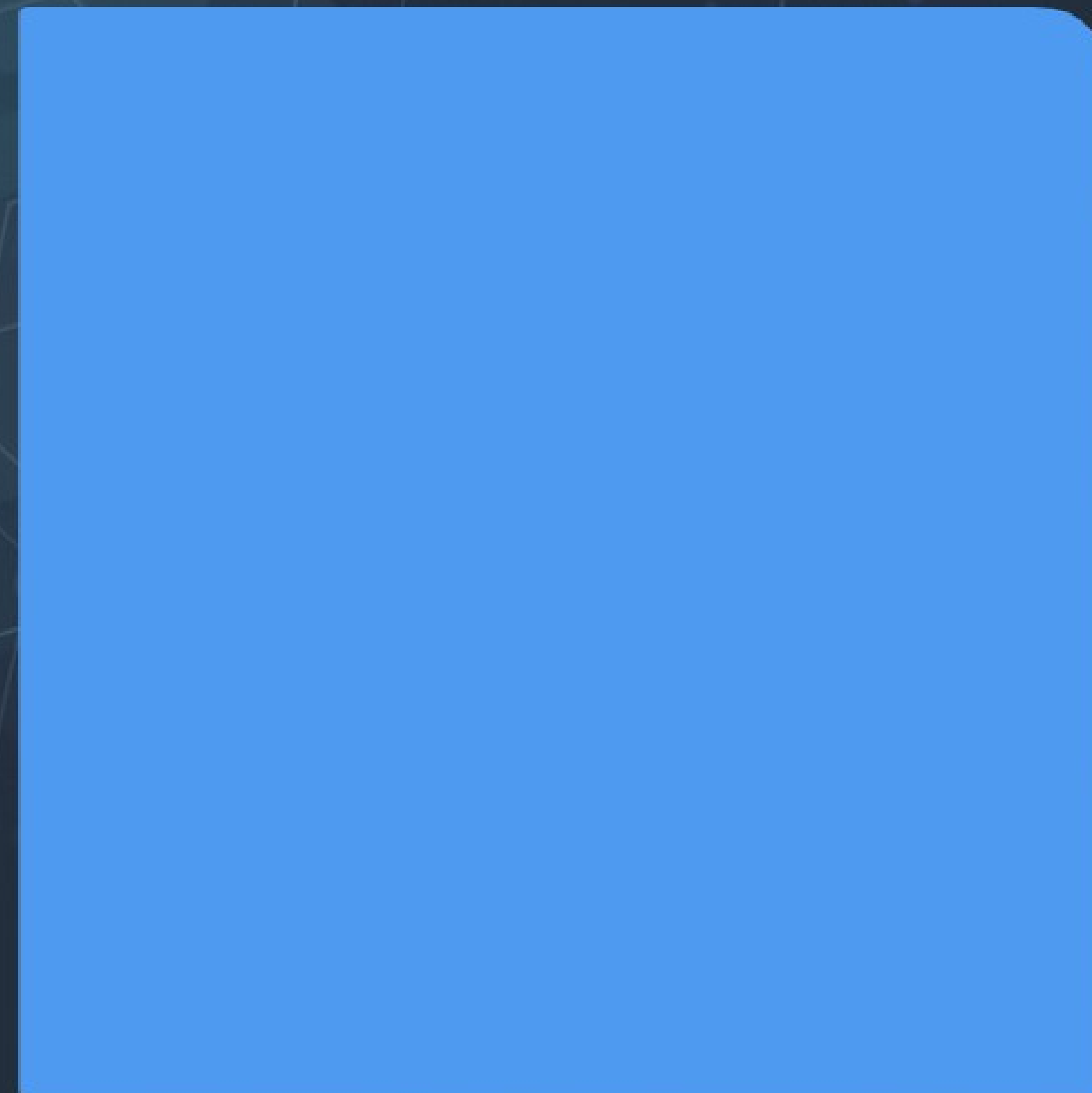
no

#1 Dates in Java 8:

- Substitute for old Java dates in java.util.Date and `java.util.Calendar`
- Thread safe
- Easy (or at least a lot easier) to work with
- Time-zone logic included
- java.time package: `LocalDate`, `LocalTime`, `LocalDateTime`

Is the *LocalDate* class thread-safe?

12 ✓



yes

0 ✗

no

0 ✗

unknown

The LocalDate class is immutable and thread-safe.

#2 Time Zones and DST (Daylight Saving Time)

- Instant: object representing a point in time
- ZonedDateTime: object representing a date and time with time zone
- ZoneId: used to convert between an Instant and a LocalDateTime, either a fixed offset or a geographical region

How do you get time in UTC standard?

9 



`Time.getUTC();`

0 

`TimeZone.getUTC();`

0 

`LocalDateTime.now();`

4 



`Instant.now();`

The *now()* method of the *Instant* class returns the current time using the best available system clock, thus converting the date and time to the UTC time zone.

#3 Clock

- Abstract class that provides access to Instant, date, and time using a time zone
- Usage is optional and allows alternate clocks
- System factory methods (.now()) rely on System.currentTimeMillis()

How do you create an instance of the Clock class in Java 8?

11 ✓



0 ✗

`Clock clock = new Clock();`

0 ✗

`Clock clock =
Clock.getTimeZone("UTC");`

0 ✗

`Clock clock = Clock.getDefault();`

To create an instance of the Clock class in Java 8, you need to use the static `systemUTC()` method from the Clock class. This method returns the current instance using the best available system clock, converting it to date and time using the UTC time zone.

#4 Built-in methods available now:

- compareTo
- isAfter
- isBefore
- isEqual

How do you check the order of two dates in Java 8?

1 ✗

Using the == operator

2 ✗

Using the equals() method

0 ✗

You cannot compare objects of
the LocalDate class in Java 8

9 ✓

Using the compareTo() method

To compare two objects of the `LocalDateTime` class in Java 8, you can use the `compareTo()` method, which returns a negative number, zero, or a positive number, depending on which object comes earlier in chronological order.

Formatting and Parsing Dates and Times

#2.1 DateTimeFormatter:

- 2022-07-08T00:11:41.262
- Java.time.format package
- Formatter is used for parsing and printing date and time objects
- Predefined formats
- Custom formats using patterns

Symbol	Represents	Explanation
Y	Year of era	Vyyy → 2022, yy → 22
D	Day of the year	A number that specifies what day of the year it is, range: 1–366
d	Day of month	A number that specifies what day of the month it is, range: 1–31
M	Month of year	M → 2, MM → 02, MMM → Feb, MMMM → February
h	Clock hour am/pm	Number representing the hour, range: 1–12
H	Hour of day	Number representing the hour, range: 0–23
m	Minute of hour	Number representing the minute of the hour range: 0–59
s	Second of minute	Number representing the second of the minute
z	Time zone name	The word representing the time zone
Z	Zone offset	Representing the offset

Most Common Letters for Patterns

Is the DateTimeFormatter class thread-safe?

0 ✓

yes

0 ✗

no

0 ✗

no idea



#2.2 Amount of Time

Period

Duration

- Represents an amount of time in seconds or nanoseconds. Minutes and hours and the DAYS unit can be used to work with Durations.

How is a *Duration* object created in Java 8? *Duration d = ...*

0 ✓

= Duration.ofHours(1)

9 ✗

= Duration.ofMinutes(60)

0 ✓

= new Duration("PT1H")

1 ✓

= Duration.parse("PT1H")

In Java 8, an object of the Duration class can be created using the static methods ofHours(), ofMinutes(), parse(), etc. You cannot create an object of the Duration class using a constructor.

#2.3 Altering Dates and Times

- Date Time are Immutable classes
- Doing calculations with the plus and minus methods
- Using Duration and Period to do calculations
- The with() and with...() methods to get a new object with an updated value

What method should be used to decrease the time by 1 hour in the LocalTime object?

10 ✓



minusHours(1)

0 ✗

subtractHours (1)

0 ✗

reduceHours(1)

1 ✗

decrementHours(1)



Quiz time

QUIZ

What does LocalDate represent?

14 ✓



3 ✗



A specific timezone

0 ✗

Timeline of events

0 ✗

Formatted time

What ZoneId in Java is used for?

22 ✓



Handling time zones

1 ✗

Manipulating strings

0 ✗

Creating timelines

0 ✗

Parsing global dates

Example: What does this code return? LocalTime.now();

2 ✗

A specific date

1 ✗

A formatted string

21 ✓

Current time without timezone

0 ✗

A timeline object

What does Instant represent?

22 ✓



A moment on the
timeline

0 ✗

Formatted date

1 ✗

A LocalDate object

0 ✗

Specific time intervals

How do you parse dates in Java?

1 ✗

Using `LocalTime.now`

2 ✗

Calling a static string method

20 ✓

Using
`DateTimeFormatter.parse()`

0 ✗

Using a timeline constructor

What does ZonedDateTime represent?

1 ✗

A formatted time zone

0 ✓

Universal time format

23 ✗

**Date and time in a specific
timezone

0 ✗

Formatted string output

What does ZoneId.systemDefault() return?

1 ✗

A formatted date

2 ✗

A timezone-aware timeline

19 ✓

The default system
timezone

0 ✗

Specific date intervals

How can you add 5 days to LocalDate?

22 ✓



Use the `plusDays(5)` method

0 ✗

Call print function

0 ✗

Direct modify date instance

0 ✗

Handle zone transitions

Which package does LocalDate belong to in Java?

21 ✓



1 ✗

java.parse.date

0 ✗

java.timezone

1 ✗

java.local.time

What does Duration measure?

22 ✓



A time span between two instants

0 ✗

System events

0 ✗

Form of timestamps

0 ✗

A zone difference

Quiz leaderboard



Q&A part

0 questions
0 upvotes

